

Physics 7B — Midterm 1 Review Worksheet

Berkeley (Physics 7B)

Topics for Midterm 1 review:

- Internal energy
- Thermal expansion
- Calorimetry
- Ideal gas law
- Heat transfer
- First law of thermodynamics
- Heat engines and efficiency
- Entropy and the second law

Exercises

Exercise 1. Gas mixing and partial pressures

[Ideal gas law]

A 6.00 L container of neon gas at a pressure of 800 torr is connected to a 10.0 L container of argon gas at a pressure of 400 torr so that the gases are allowed to mix throughout both containers.

- Calculate the total pressure of the gas after mixing.
- Calculate the mole fraction of each gas after mixing.
- If both gases are forced together into the 10.00 L container, determine the partial pressure of each gas in the 10.00 L container.

Exercise 2. A three-process cycle

[Ideal gas law] [First law] [Entropy / Second Law]

An ideal gas starts out in a state A, with pressure P_A , volume V_A , and temperature T_A . First, the gas expands isothermally until its volume has tripled and it reaches point B. Then, the gas cools isovolumetrically to reach point C, at which point it is adiabatically compressed to return to point A. This gas has $\gamma = \frac{C_P}{C_V} = \frac{5}{3}$. (It will be necessary to use the following relation for an adiabatic process on an ideal gas: $TV^{\gamma-1} = \text{constant}$.)

- (a) Draw a P - V diagram for this cycle.
- (b) What is the work done by the gas during each step of this cycle?
- (c) What is the heat flowing into the gas at each step of this cycle?
- (d) What is the change in entropy in the gas during each step of this cycle?

Exercise 3. Conduction through a cylinder

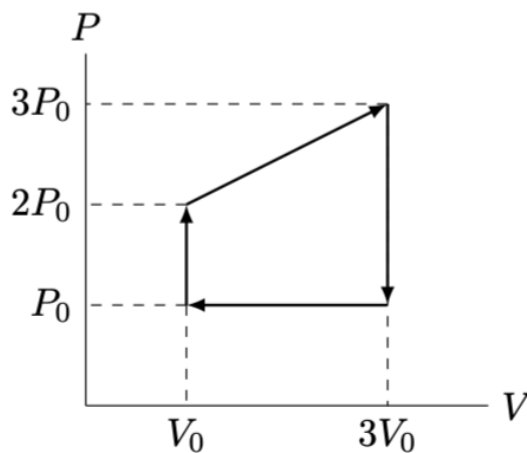
[Heat transfer]

Consider the cross-sectional slice of a cylinder (a cylindrical shell) with inner radius r_1 and outer radius r_2 . Assume that the cylinder has height h and is made of a homogeneous material with thermal conductivity k . The inner wall is kept at temperature T_1 , and the outer wall is kept at temperature $T_2 < T_1$. The system has reached a steady state of heat transfer; i.e., the rate of heat flow is constant: $\frac{dQ}{dt} = q$.

- (a) Use the conductive heat transfer equation to find an expression for the temperature gradient. Integrate to find a formula for the temperature T as a function of the radius r .
- (b) Use your formula from part (a) to solve for q . (Hint: when you evaluate T at the inner and outer walls of the cylinder, what should you get?)

Exercise 4. Heat engine cycle on a P - V diagram [Heat engines and efficiency] [First law] [Internal energy] [Ideal gas law]

Consider the heat engine cycle shown on the following P - V diagram. Assuming that the gas is ideal and monatomic, calculate the efficiency of this engine, and compare it to the theoretical maximum efficiency.



Exercise 5. Ice–water calorimetry in an isolated container
[Internal energy]

[Calorimetry] [First law]

A container, thermodynamically isolated, contains a mass M of ice at temperature $T = 0^\circ\text{C}$. An equivalent mass M of water is now added to the container. The temperature of the water is T_W . The pressure of the water–ice system is P , the specific heat of water is C_W , and the latent heat of fusion of the ice is L . The total volume of the water and ice system does not change when the ice melts into water. Neglect the heat capacity of the container. Find:

- (a) The variation of internal energy of the system of water and ice, between the initial and final state.
- (b) If all the ice melts, find the final equilibrium temperature of the water and ice system.

Exercise 6. Calorimetry with container + liquid + hot pellet
[Internal energy]

[Calorimetry] [First law]

A cubic iron container with mass $M = 0.5\text{ kg}$ and side length $L = 1\text{ m}$ is half full of a strange liquid with density $\rho = 300\text{ kg m}^{-3}$. The system is originally at temperature $T_0 = 20^\circ\text{C}$. We then drop a very hot aluminum pellet of mass $m = 0.1\text{ kg}$ and temperature $T_{\text{Al},0} = 200^\circ\text{C}$ into the liquid. The specific heats of iron, aluminum, and the liquid are $c_{\text{Fe}} = 450\text{ J kg}^{-1}\text{ K}^{-1}$, $c_{\text{Al}} = 900\text{ J kg}^{-1}\text{ K}^{-1}$, and $c_{\text{liq}} = 750\text{ J kg}^{-1}\text{ K}^{-1}$, respectively. Calculate the final equilibrium temperature T_f of the system (container + liquid + pellet). Neglect any work done and assume no heat exchange with the environment.

Exercise 7. Campfire skewer + water (entropy production)
[Second Law]

[Calorimetry] [Entropy /

At a campfire high in the Sierra, you are roasting marshmallows using a metal skewer of mass M_S and specific heat c_S . The skewer is at temperature T_H . You dip the skewer into a pan filled with a mass M_W of water (specific heat c_W) at temperature T_C until the skewer and the water reach thermal equilibrium (ignore the water container, and assume there is no water evaporation).

- (a) What is the thermal equilibrium temperature T_F ?
- (b) What is the change in entropy of the skewer ΔS_S ?
- (c) What is the change in entropy of the water ΔS_W ?
- (d) What is the total change in entropy ΔS_{tot} for the system (water + skewer)?
- (e) Quoting a thermodynamic law, explain and deduce the sign of ΔS_{tot} .

Exercise 8. Otto cycle [Engines, efficiency] [First law] [Entropy / Second Law] [Ideal gas law]

The Otto cycle comprises four segments: (1) adiabatic expansion from $A \rightarrow B$, (2) isovolumetric cooling from $B \rightarrow C$, (3) adiabatic compression from $C \rightarrow D$, and (4) isovolumetric heating from $D \rightarrow A$.

- (a) Sketch the P - V diagram.
- (b) What is the heat absorbed in each segment in terms of P and V ?
- (c) Calculate the efficiency $e = W/Q_H$ of this engine for an ideal diatomic gas, where W is the work done *by* the gas in a full cycle and Q_H is the total heat flowing into the gas. Compare this to the efficiency of the Carnot cycle.
- (d) Would the efficiency increase or decrease for a monoatomic gas? Show the comparison explicitly.
- (e) Calculate the change in entropy across the whole cycle.

Solutions

Exercise 1. Gas mixing and partial pressures

[Ideal gas law]

Assume the two gases are ideal and the mixing process does not change the temperature.

Let container 1 (neon) have $(P_1, V_1) = (800 \text{ torr}, 6.00 \text{ L})$ and container 2 (argon) have $(P_2, V_2) = (400 \text{ torr}, 10.0 \text{ L})$.

(a) Using $n_i = \frac{P_i V_i}{RT}$ and $P_f(V_1 + V_2) = (n_1 + n_2)RT$, we get

$$P_f = \frac{P_1 V_1 + P_2 V_2}{V_1 + V_2} = \frac{(800)(6.00) + (400)(10.0)}{6.00 + 10.0} \text{ torr} = 550 \text{ torr}.$$

(b) Mole fractions:

$$x_{\text{Ne}} = \frac{n_1}{n_1 + n_2} = \frac{P_1 V_1}{P_1 V_1 + P_2 V_2} = \frac{4800}{4800 + 4000} \approx 0.545, \quad x_{\text{Ar}} = \frac{P_2 V_2}{P_1 V_1 + P_2 V_2} = \frac{4000}{8800} \approx 0.455.$$

(c) If the mixture is compressed into $V = 10.00 \text{ L}$ at the same temperature, the total pressure is

$$P = \frac{(n_1 + n_2)RT}{V} = \frac{P_1 V_1 + P_2 V_2}{V} = \frac{(800)(6.00) + (400)(10.0)}{10.00} \text{ torr} = 880 \text{ torr}.$$

The partial pressures are $p_i = x_i P$:

$$p_{\text{Ne}} = (0.545)(880) \approx 480 \text{ torr}, \quad p_{\text{Ar}} = (0.455)(880) \approx 400 \text{ torr}.$$

Exercise 2. A three-process cycle

[Ideal gas law] [First law] [Entropy / Second Law]

Let n be the number of moles. Along $A \rightarrow B$ the temperature is constant $T = T_A$, and the volume changes from V_A to $V_B = 3V_A$. Along $B \rightarrow C$ the volume is constant $V = 3V_A$. Along $C \rightarrow A$ the process is adiabatic with $TV^{\gamma-1} = \text{const}$ and $\gamma = 5/3$.

(a) The P - V diagram has:

- $A \rightarrow B$: isothermal expansion (a decreasing hyperbola) from (V_A, P_A) to $(3V_A, P_B)$.
- $B \rightarrow C$: isochoric cooling (vertical line down at $V = 3V_A$).
- $C \rightarrow A$: adiabatic compression (steeper curve than the isotherm) returning to (V_A, P_A) .

From the isothermal step, $P_B = P_A/3$.

(b) Work by the gas:

$$W_{AB} = \int_{V_A}^{3V_A} p dV = nRT_A \int_{V_A}^{3V_A} \frac{dV}{V} = nRT_A \ln 3 = P_A V_A \ln 3.$$

$$W_{BC} = 0 \quad (\text{constant } V).$$

For the adiabatic compression, $Q_{CA} = 0$ so $W_{CA} = -\Delta U_{CA} = -nC_V(T_A - T_C)$. Using $TV^{\gamma-1} = \text{const}$ between C and A :

$$T_C(3V_A)^{\gamma-1} = T_A(V_A)^{\gamma-1} \Rightarrow T_C = T_A 3^{-(\gamma-1)} = T_A 3^{-2/3}.$$

With $\gamma = 5/3$ we have $C_V = \frac{R}{\gamma-1} = \frac{3}{2}R$, hence

$$W_{CA} = -nC_V(T_A - T_A 3^{-2/3}) = -\frac{3}{2}nRT_A(1 - 3^{-2/3}) = \frac{3}{2}P_A V_A(3^{-2/3} - 1).$$

(c) Heat into the gas:

$$Q_{AB} = \Delta U_{AB} + W_{AB} = 0 + W_{AB} = P_A V_A \ln 3.$$

$$Q_{BC} = \Delta U_{BC} = nC_V(T_C - T_B) = \frac{3}{2}nR(T_A 3^{-2/3} - T_A) = \frac{3}{2}P_A V_A(3^{-2/3} - 1).$$

$$Q_{CA} = 0 \quad (\text{adiabatic}).$$

(d) Entropy changes (assuming reversibility):

$$\Delta S_{AB} = nR \ln\left(\frac{V_B}{V_A}\right) = nR \ln 3.$$

Along $B \rightarrow C$ (constant V):

$$\Delta S_{BC} = nC_V \ln\left(\frac{T_C}{T_B}\right) = \frac{3}{2}nR \ln(3^{-2/3}) = -nR \ln 3.$$

Along the reversible adiabatic step, $\Delta S_{CA} = 0$. The net entropy change over the full cycle is 0, as expected for a reversible cycle that returns to its initial state.

Exercise 3. Conduction through a cylinder

[Heat transfer]

In steady state, Fourier's law gives the heat current

$$q = \frac{dQ}{dt} = -kA(r) \frac{dT}{dr}, \quad A(r) = 2\pi r h.$$

Therefore

$$q = -2\pi k h r \frac{dT}{dr} \Rightarrow \frac{dT}{dr} = -\frac{q}{2\pi k h} \frac{1}{r}.$$

(a) Integrating,

$$T(r) = -\frac{q}{2\pi k h} \ln r + C,$$

where C is a constant of integration.

(b) Apply the boundary conditions $T(r_1) = T_1$ and $T(r_2) = T_2$:

$$T_1 = -\frac{q}{2\pi kh} \ln r_1 + C, \quad T_2 = -\frac{q}{2\pi kh} \ln r_2 + C.$$

Subtract to eliminate C :

$$T_1 - T_2 = \frac{q}{2\pi kh} \ln\left(\frac{r_2}{r_1}\right) \quad \Rightarrow \quad q = \frac{2\pi kh (T_1 - T_2)}{\ln\left(\frac{r_2}{r_1}\right)}.$$

Exercise 4. Heat engine cycle on a P - V diagram [*Heat engines and efficiency*] [*First law*] [*Internal energy*] [*Ideal gas law*]

Label the corners of the cycle a, b, c, d starting at the lower-left corner (P_0, V_0) and moving clockwise. Assume the gas is monatomic, so $\Delta U = \frac{3}{2}nR\Delta T$.

Using $P_0V_0 = nRT_0$ at point a , the ideal gas law gives

$$T_b = 2T_0, \quad T_c = 9T_0, \quad T_d = 3T_0.$$

The operating temperature range is $[T_0, 9T_0]$, so the Carnot bound is

$$\eta_{\max} = 1 - \frac{T_{\text{cold}}}{T_{\text{hot}}} = 1 - \frac{T_0}{9T_0} = \frac{8}{9}.$$

Compute heat and work on each leg (with $W > 0$ done by the gas).

- $a \rightarrow b$ (isochoric): $W_{ab} = 0$, so

$$Q_{ab} = \Delta U_{ab} = \frac{3}{2}nR(T_b - T_a) = \frac{3}{2}nRT_0 = \frac{3}{2}P_0V_0.$$

- $b \rightarrow c$ (linear expansion): work is the trapezoid area under the line,

$$W_{bc} = \frac{1}{2}(P_b + P_c)(V_c - V_b) = \frac{1}{2}(2P_0 + 3P_0)(2V_0) = 5P_0V_0.$$

Also

$$\Delta U_{bc} = \frac{3}{2}nR(T_c - T_b) = \frac{3}{2}nR(7T_0) = \frac{21}{2}P_0V_0.$$

so

$$Q_{bc} = \Delta U_{bc} + W_{bc} = \frac{31}{2}P_0V_0.$$

- $c \rightarrow d$ (isochoric): $W_{cd} = 0$, and

$$\Delta U_{cd} = \frac{3}{2}nR(T_d - T_c) = \frac{3}{2}nR(-6T_0) = -9P_0V_0.$$

so $Q_{cd} = -9P_0V_0$.

- $d \rightarrow a$ (isobaric compression at P_0): $W_{da} = P_0(V_a - V_d) = P_0(V_0 - 3V_0) = -2P_0V_0$. Also

$$\Delta U_{da} = \frac{3}{2}nR(T_a - T_d) = \frac{3}{2}nR(-2T_0) = -3P_0V_0.$$

so

$$Q_{da} = \Delta U_{da} + W_{da} = -5P_0V_0.$$

Net work and heat input:

$$W_{\text{net}} = W_{ab} + W_{bc} + W_{cd} + W_{da} = 0 + 5 + 0 - 2 = 3P_0V_0,$$

$$Q_{\text{in}} = Q_{ab} + Q_{bc} = \frac{3}{2}P_0V_0 + \frac{31}{2}P_0V_0 = 17P_0V_0.$$

Thus the efficiency is

$$\eta = \frac{W_{\text{net}}}{Q_{\text{in}}} = \frac{3}{17},$$

which is much less than the Carnot bound $\eta_{\text{max}} = 8/9$.

Exercise 5. Ice–water calorimetry in an isolated container

[Calorimetry] [First law]

[Internal energy]

- (a) The system is thermodynamically isolated, so the net heat exchange with the environment is zero and the total work done on/by the system is zero (no volume change overall). Thus the total change in internal energy of the combined (ice+water) system is

$$\Delta U_{\text{total}} = 0.$$

- (b) Set the net heat exchanged between components to zero:

$$Q_{\text{ice}} + Q_W = 0.$$

Taking the ice initially at 0°C , if all the ice melts and the final temperature is $T_f > 0^\circ\text{C}$, then

$$Q_{\text{ice}} = ML + MC_W(T_f - 0), \quad Q_W = MC_W(T_f - T_W).$$

So

$$ML + MC_WT_f + MC_W(T_f - T_W) = 0 \Rightarrow 2MC_WT_f = MC_WT_W - ML,$$

and therefore

$$T_f = \frac{1}{2C_W} (C_WT_W - L).$$

(This result applies under the stated condition that all the ice melts; equivalently $T_f \geq 0$.)

Exercise 6. Calorimetry with container + liquid + hot pellet

[Calorimetry] [First law]

[Internal energy]

Assume the container, liquid, and pellet come to a common final temperature T_f and the system is isolated, so the net change in internal energy is zero:

$$0 = \Delta U_{\text{liq}} + \Delta U_{\text{Fe}} + \Delta U_{\text{Al}}.$$

The liquid volume is half the cube: $V_{\text{liq}} = \frac{1}{2}L^3 = \frac{1}{2}\text{ m}^3$, so its mass is

$$m_{\text{liq}} = \rho V_{\text{liq}} = 300 \cdot \frac{1}{2} = 150 \text{ kg}.$$

Using $\Delta U = mc \Delta T$ for each component:

$$0 = m_{\text{liq}}c_{\text{liq}}(T_f - T_0) + Mc_{\text{Fe}}(T_f - T_0) + mc_{\text{Al}}(T_f - T_{\text{Al},0}).$$

Solve for T_f :

$$T_f = \frac{(m_{\text{liq}}c_{\text{liq}} + Mc_{\text{Fe}})T_0 + mc_{\text{Al}}T_{\text{Al},0}}{m_{\text{liq}}c_{\text{liq}} + Mc_{\text{Fe}} + mc_{\text{Al}}}.$$

Plugging in numbers ($m_{\text{liq}}c_{\text{liq}} = 150 \cdot 750 = 112,500$, $Mc_{\text{Fe}} = 0.5 \cdot 450 = 225$, $mc_{\text{Al}} = 0.1 \cdot 900 = 90$ in J/K):

$$T_f = \frac{(112,500 + 225)(20) + (90)(200)}{112,500 + 225 + 90} ^\circ\text{C} \approx 20.14 ^\circ\text{C}.$$

Exercise 7. Campfire skewer + water (entropy production) *[Calorimetry] [Entropy / Second Law]*

(a) Energy conservation (heat lost by skewer = heat gained by water):

$$M_S c_S (T_H - T_F) = M_W c_W (T_F - T_C) \quad \Rightarrow \quad T_F = \frac{M_S c_S T_H + M_W c_W T_C}{M_S c_S + M_W c_W}.$$

(b) Entropy change of the skewer (take c_S constant):

$$\Delta S_S = \int_{T_H}^{T_F} \frac{dQ_{\text{rev}}}{T} = \int_{T_H}^{T_F} \frac{M_S c_S dT}{T} = M_S c_S \ln\left(\frac{T_F}{T_H}\right) < 0,$$

since $T_F < T_H$.

(c) Entropy change of the water:

$$\Delta S_W = \int_{T_C}^{T_F} \frac{M_W c_W dT}{T} = M_W c_W \ln\left(\frac{T_F}{T_C}\right) > 0,$$

since $T_F > T_C$.

(d) Total entropy change of the system:

$$\Delta S_{\text{tot}} = \Delta S_S + \Delta S_W = M_S c_S \ln\left(\frac{T_F}{T_H}\right) + M_W c_W \ln\left(\frac{T_F}{T_C}\right).$$

(e) By the Second Law of Thermodynamics, in an isolated system an irreversible process (heat flow from hot to cold) increases the total entropy, so $\Delta S_{\text{tot}} > 0$.

Exercise 8. Otto cycle [Engines, efficiency] [First law] [Entropy / Second Law] [Ideal gas law]

- (a) The P - V diagram consists of two adiabatic curves connecting volumes V_A and V_B , and two vertical (constant-volume) legs at $V = V_A$ and $V = V_B$: $A \rightarrow B$ adiabatic expansion, $B \rightarrow C$ isochoric cooling at V_B , $C \rightarrow D$ adiabatic compression, and $D \rightarrow A$ isochoric heating at V_A .
- (b) For the adiabatic legs, $Q_1 = Q_{AB} = 0$ and $Q_3 = Q_{CD} = 0$. For the isochoric legs, $W = 0$ so $Q = \Delta U = nC_V\Delta T$. Using $PV = nRT$ at fixed volume:

$$Q_2 = Q_{BC} = nC_V(T_C - T_B) = \frac{C_V}{R}(P_C - P_B)V_B, \quad Q_4 = Q_{DA} = nC_V(T_A - T_D) = \frac{C_V}{R}(P_A - P_D)V_A.$$

For a diatomic ideal gas (no vibrational modes), $C_V = \frac{5}{2}R$, so

$$Q_2 = \frac{5}{2}(P_C - P_B)V_B, \quad Q_4 = \frac{5}{2}(P_A - P_D)V_A.$$

- (c) Over a full cycle, $\Delta U = 0$, so $W_{\text{net}} = Q_{\text{net}} = Q_H - Q_C$. Heat input occurs on $D \rightarrow A$, so $Q_H = Q_4$ (positive), and heat rejected occurs on $B \rightarrow C$, so $Q_C = -Q_2$ (positive magnitude $|Q_2|$). Hence

$$e_{\text{Otto}} = \frac{W_{\text{net}}}{Q_H} = 1 - \frac{Q_C}{Q_H} = 1 - \frac{|Q_2|}{|Q_4|} = 1 - \frac{(P_B - P_C)V_B}{(P_A - P_D)V_A}.$$

The Carnot efficiency between the hottest and coldest temperatures is

$$e_{\text{Carnot}} = 1 - \frac{T_C}{T_A} = 1 - \frac{P_C V_B}{P_A V_A}.$$

Using adiabatic relations $P_A V_A^\gamma = P_B V_B^\gamma$ and $P_D V_A^\gamma = P_C V_B^\gamma$ gives $P_D/P_A = P_C/P_B$, which simplifies the Otto efficiency to

$$e_{\text{Otto}} = 1 - \frac{P_B V_B}{P_A V_A}.$$

Since $P_B > P_C$, one finds $e_{\text{Otto}} < e_{\text{Carnot}}$ for the same temperature extremes.

- (d) From $P_A V_A^\gamma = P_B V_B^\gamma$ we have $\frac{P_B}{P_A} = \left(\frac{V_A}{V_B}\right)^\gamma$, and thus

$$e_{\text{Otto}} = 1 - \left(\frac{V_A}{V_B}\right)^{\gamma-1}.$$

For a monatomic gas $\gamma = 5/3$ while for a diatomic gas $\gamma = 7/5$. Since $V_B > V_A$, a larger γ makes $\left(\frac{V_A}{V_B}\right)^{\gamma-1}$ smaller, so the efficiency is higher for a monatomic gas.

- (e) Over a full cycle, the entropy change is zero: $\Delta S_{\text{cycle}} = 0$ (state function). More explicitly, ΔS is zero on the reversible adiabatic legs, and on the isochoric legs

$$\Delta S_{BC} = nC_V \ln\left(\frac{T_C}{T_B}\right) = nC_V \ln\left(\frac{P_C}{P_B}\right), \quad \Delta S_{DA} = nC_V \ln\left(\frac{T_A}{T_D}\right) = nC_V \ln\left(\frac{P_A}{P_D}\right),$$

and using $P_D/P_A = P_C/P_B$ gives $\Delta S_{BC} + \Delta S_{DA} = 0$.